

LINA GOTZHA

RESEARCH SCIENTIST/ENGINEER

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Research Scientist with a strong foundation in agile software development, machine learning, and data analysis. Currently developing an interpretable tool to support neuroscience labs focusing on cognitive processes research. Previously at AT&T, advocated for responsible and explainable AI in predictive systems. Passionate about modeling human-environment interactions using transparent, interpretable methods that reveal complex patterns in data.

Independent Research

2023-now

I use LLMs as cognitive mirrors, applying representation tilts and adapters to probe how memory alters narrative meaning, grounded in narrative fMRI datasets and Brain-LLM research.

Professional Experience

Data Scientist, Neuroscience Lab, Haifa University, Isarel

11.2022 - 5.2023

- Analyzed electrophysiological data from rats models using **machine learning** to classify sociability traits relevant to autism research.
- Building a proof-of-concept for an interpretable, human-centered AI framework to support neuroscience research, with results to be published.

Data Scientist, AT&T

2019-2022

Playing a pivotal role in developing cutting-edge AI solutions and optimizing **data analytics** pipelines to enhance operational efficiency and customer service.

- Led project Profx, an **AI-driven** interactive platform promoting explainable and responsible AI within data science workflows, enhancing data scientists' decision making. Responsible for the data science core component of the project, designing, verifying, and implementing the learning pipeline to ensure a robust and interpretable process.
- Improved runtime efficiency by 50% in a predictive issue ticketing machine learning pipeline, by leveraging **Spark** for distributed processing and **HDFS** for scalable data storage, to classify service tickets and predict delivery dates, boosting performance across AT&T services.
- Designed and implemented a **Python**-based comparative testing tool for large-scale data validation in a **Dockerized**, high-performance distributed system, enabling detection of misplaced, missing, and corrupted files across a data management platform.

Java Developer, Amdocs

2016-2017, 2012-2015

- Led end-to-end design and implementation of features for application development across both **Windows** and **Linux** environments. Developed logic and tests using **Java**, while utilizing J2EE for enhanced functionality in scale.
- Recognized for mentoring team members, leading knowledge transfer sessions, delivering on-site customer support, and promoting collaboration with international teams to ensure the smooth development and delivery of high-quality solutions.

Education

M. Sc. in Computer Science, Oxford University, UK

2017 - 2018

Research thesis in automated strategy synthesis for condition-based maintenance, using parameterized Markov chains, Monte Carlo simulation and Bayesian inference. Majored in advanced machine learning, probabilistic models, AI ethics.

B. Sc. in Industrial Engineering & Management, Technion, Israel

2007-2012

Final data driven research project in collaboration with Caesarstone company. Identified that 20% of products drive 80% of revenue, optimizing data analysis and improving forecast accuracy, while implementing the Holt-Winters method of exponential smoothing.

Fluent in English, Arabic, Hebrew and Circassian